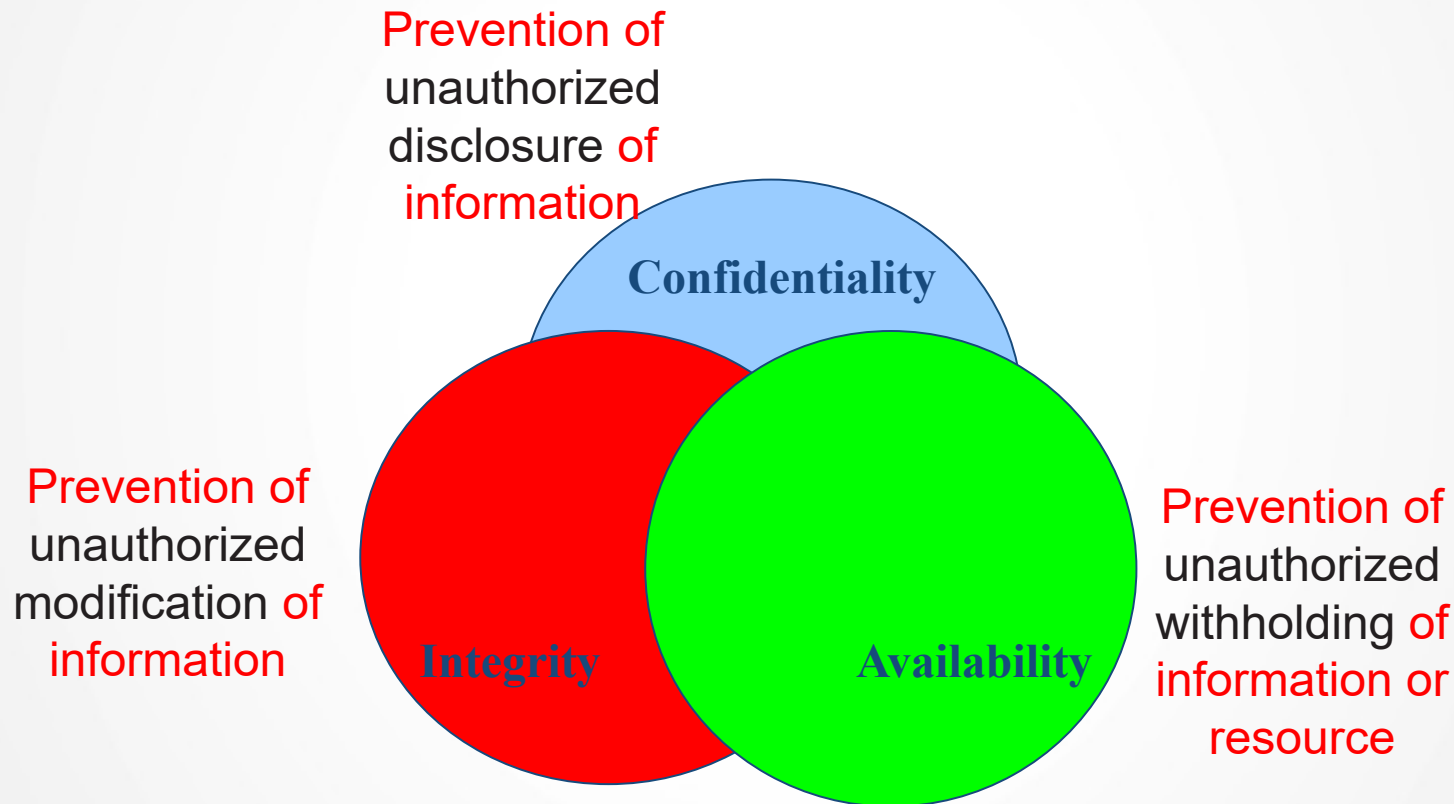


Fundamentals of Computer Security

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Computer Security and Privacy

Security Goals (Pillars) - CIA



Computer Security and Privacy

≡ **Security** in general is about **protection of assets**.

- To protect our assets, we must know the **assets and their values**.

≡ **Classification of protection measures includes**

- **Prevention**: take measures to prevent the damage
- **Detection**: when, how and who of the damage.
- **Reaction**: take measures to recover from damage.

Computer Security and Privacy

☰ **Example (1): Protecting valuable items at home from a burglar:**

- **Prevention:** Locks on the door, guards, hidden places, ...
- **Detection:** Burglar alarm, guards, CCTV, ...
- **Reaction:** Calling the police, replace the stolen item, ...

Computer Security and Privacy

Example (2): Protecting a fraudster from using our credit card in Internet purchase

- **Prevention:** Encrypt when placing order, perform some check before placing order, or don't use credit card number on Internet.
- **Detection:** A transaction that you had not authorized appears on your credit card statement.
- **Reaction:** Ask for new card, recover cost of the transaction from the insurance, the card issuer or the merchant.

Computer Security and Privacy/ Overview

Definitions

 **Security:** The protection of computer assets from unauthorized access, use, alteration, degradation, destruction, and other threats.

 **Privacy:** The right of the individual to be protected against intrusion into his/her personal life or affairs, or those of his/her family.

 **Security/Privacy Threat:** Any person, act, or object that poses a danger to computer security/privacy.

Computer Security and Privacy/ History

☞ Until 1960s computer security was limited to physical protection of computers

☞ In the 60s and 70s

● Evolutions

- Computers became interactive
- Multiuser/Multiprogramming was invented
- More and more data started to be stored in computer databases

● Organizations and individuals **started to worry** about

- What the other persons using computers are doing to their data
- What is happening to their private data stored in large databases

Computer Security and Privacy/ History

In the 80s and 90s

● Evolutions

- Personal computers were popularized
- LANs and Internet invaded the world
- Applications such as E-commerce, E-government and
- E-health started to develop
- Viruses become majors threats
- Organizations and individuals **started to worry about**
 - Who has access to their computers and data
 - Whether they can trust an e-mail, a website, etc.
 - Whether their privacy is protected in the connected world

Computer Security and Privacy/ History

☰ Famous security problems

● **Morris worm** – Internet Worm

- November 2, 1988 a worm attacked more than 60,000 computers around the USA
- The worm attacks computers, and when it has installed itself, it multiplies itself, freezing the computer
- It exploited UNIX security holes in Sendmail and Finger
- A nationwide effort enabled to solve the problem within 12 hours
- Robert Morris became **the first person** to be indicted under the **Computer Fraud** and Abuse Act.
 - He was sentenced to three years of probation, 400 hours of community service and a fine of \$10,500
- Until recently, he has been an **associate professor** at the Massachusetts Institute of Technology (MIT)

Computer Security and Privacy/ History

☰ Famous security problems ...

● NASA shutdown

- In 1990, an Australian computer science student was charged for shutting down NASA's computer system for 24 hours

● Airline computers

- In 1998, a major travel agency discovered that someone penetrated its ticketing system and has printed airline tickets illegally

● Bank theft

- In 1984, a bank manager was able to steal **\$25 million** through un-audited computer transactions

Computer Security and Privacy/ History

☰ Famous security problems ...

☰ In **Ethiopia**

- Employees of a company managed to **change their salaries** by fraudulently modifying the company's database
- In 1990s Internet password theft
 - *Hundreds of dial-up passwords were stolen and sold to other users*
 - *Many of the owners lost tens of thousands of Birr each*
- A major company suspended the use of a **remote login**

☰ In **Africa: Cote d'Ivoire**

- An employee who has been fired by his company deleted all the data in his company's computer

Computer Security and Privacy/ History

Early Efforts

- 1960s: Marked as the beginning of true computer security

● 1970s: Tiger teams

- Government and industry sponsored crackers who attempted to break down defenses of computer systems in order to uncover vulnerabilities so that patches can be developed

● 1970s: Research and modeling

- Identifying security requirements
- Formulating security policy models
- Defining guidelines and controls
- Development of secure systems

Computer Security and Privacy/ Legal Issues

≡ In the US, legislation was enacted with regards to computer security and privacy starting from late **1960s**.

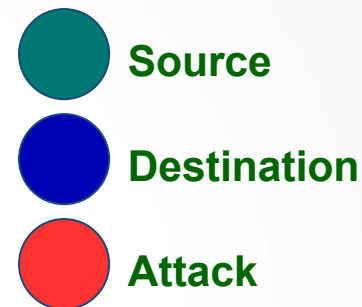
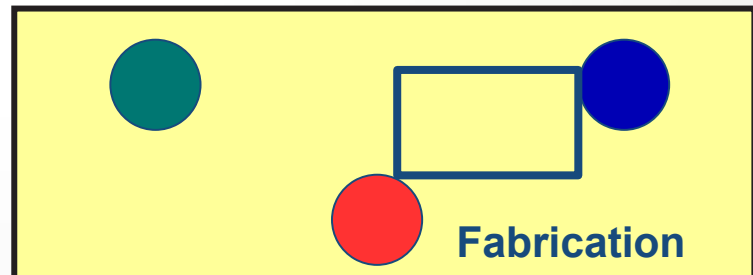
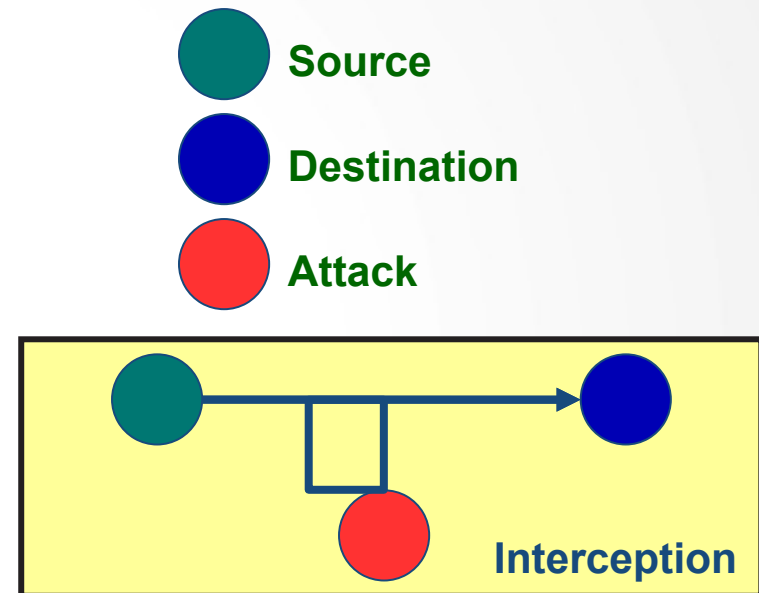
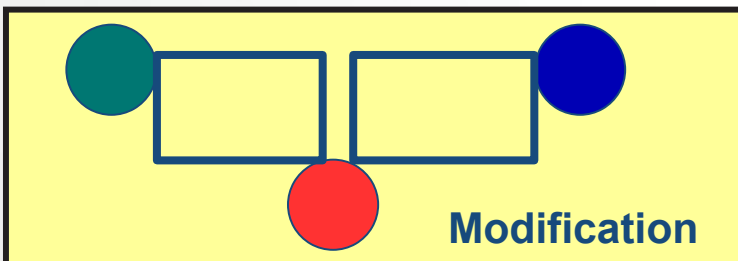
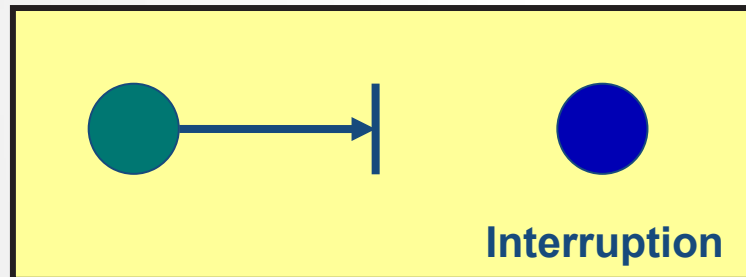
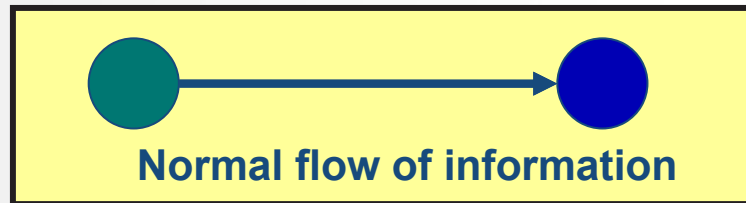
≡ European Council adopted a convention on Cyber-crime in **2001**.

≡ The World Summit for Information Society considered computer security and privacy as a subject of discussion in **2003 and 2005**.

≡ The Ethiopian Penal Code of **2005** has articles on data and computer related crimes.

Computer Security and Privacy/Attacks

Categories of Attacks/Threats (W. Stallings)



Computer Security and Privacy/ Physical Security

“The most robustly secured computer that is left sitting unattended in an unlocked room is not at all secure !!”

[Chuck Easttom]

Computer Security and Privacy/ Physical Security

☰ Physical security is the use of **physical controls** to protect premises, site, facility, building or other physical asset of an organization [Lawrence Fennelly]

☰ Physical security protects your **physical computer facility** (your building, your computer room, your computer, your disks and other media) [Chuck Easttom].

=>

Physical security is much more difficult to achieve today than some decades ago

Threats and vulnerabilities

Natural Disasters

- Fire and smoke

- Fire can occur anywhere

- Solution – Minimize risk

- *Good policies: NO SMOKING, etc..*

- *Fire extinguisher, good procedure and training*

- *Fireproof cases (and other techniques) for backup tapes*

- *Fireproof doors*

- Climate

- Heat

- Direct sun

- Humidity

Computer Security and Privacy/ Physical Security

Threats and vulnerabilities ...

Natural Disasters ...

- Hurricane, storm, cyclone
- Earthquakes
- Water
 - Flooding can occur even when a water tap is not properly closed
- Electric supply
 - Voltage fluctuation
 - *Solution: Voltage regulator*
- Lightning

 **Avoid having servers in areas often hit by Natural Disasters!**

Threats and vulnerabilities ...

People

- Intruders

- Thieves

- People who have been given access unintentionally by the insiders

- Employees, contractors, etc. who have access to the facilities

- External thieves

- Portable computing devices can be stolen outside the organization's premises

Loss of a computing device

- Mainly laptop

Safe area

- Safe area often is a locked place where only authorized personnel can have access
- Organizations usually have safe area for keeping computers and related devices

Computer Security and Privacy/ Physical Security

Safe area ... Challenges

☰ Is the area **inaccessible** through other openings (window, roof-ceilings, ventilation hole, etc.)?

- Design of the building with security in mind
- Know the architecture of your building

☰ **During opening hours**, is it always possible to detect when unauthorized person tries to get to the safe area?

- Surveillance/guards, video-surveillance, automatic-doors with security code locks, alarms, etc.
- Put signs so that everybody sees the safe area

Computer Security and Privacy/ Physical Security

Safe area...Locks

☐ Are the locks reliable?

- The effectiveness of locks depends on the design, manufacture, installation and maintenance of the keys!
- Among the attacks on locks are:
 - Illicit keys
 - *Duplicate keys*
 - Avoid access to the key by unauthorized persons even for a few seconds
 - Change locks/keys frequently
 - Key management procedure
 - *Lost keys*
 - Notify responsible person when a key is lost
 - There should be no label on keys
 - Circumventing of the internal barriers of the lock
 - *Directly operating the bolt completely bypassing the locking mechanism which remains locked*
 - Forceful attacks:
 - *Punching, Drilling, Hammering, etc.*

Safe area... Surveillance with Guards

☰ Surveillance with guards

- The most common in Ethiopia
- Not always the most reliable since it adds a lot of human factor
- Not always practical for users (employees don't like to be questioned by guards wherever they go)

Computer Security and Privacy/ Physical Security

Safe area... Surveillance with Video

Surveillance with video

- Uses Closed Circuit Television (CCTV)
- Started in the 1960s
- Become more and more popular with the worldwide increase of theft and terrorism
- Advantages
 - A single person can monitor more than one location
 - The intruder doesn't see the security personnel
 - It is cheaper after the initial investment
 - It can be recorded and be used for investigation
 - Since it can be recorded the security personnel is more careful
 - Today's digital video-surveillance can use advanced techniques such as face recognition to detect terrorists, wanted people, etc.
- Drawback
 - Privacy concerns

Internal Human factor - Personnel

- ≡ Choose employees carefully
 - Personal integrity should be as important a factor in the hiring process as technical skills
- ≡ Create an atmosphere in which the levels of employee loyalty, morale, and job satisfaction are high
- ≡ Remind employees, on a regular basis, of their continuous responsibilities to protect the organization's information

Internal Human factor – Personnel ...

- ≡ Establish procedures for proper destruction and disposal of obsolete programs, reports, and data
- ≡ Act defensively when an employee must be discharged, either for cause or as part of a cost reduction program
 - Such an employee should not be allowed access to the system and should be carefully watched until he or she leaves the premises
 - Any passwords used by the former employee should be immediately disabled

Computer Security Attacks and Threats

Computer security/ Attacks & Threats

≡ A computer **security threat** is any person, act, or object that poses a danger to computer security

≡ Computer world is full of threats!

≡ And so is the real world!

● Thieves, pick-pockets, burglars, murderers, drunk drivers, ...

Computer security/ Attacks & Threats

What is the right attitude?

- To do what you do in real life

What do you do in real life?

- You learn about the threats
 - What are the threats
 - How can these threats affect you
 - What is the risk for you to be attacked by these threats
 - How you can protect yourself from these risks
 - How much does the protection cost
 - What you can do to limit the damage in case you are attacked
 - How you can recover in case you are attacked
- Then, you protect yourself in order to limit the risk but to continue to live your life

 You need to do exactly the same thing with computers!

Computer security/ Attacks & Threats

Types of Threats/Attacks ... (Chuck Eastom)

Physical Attack

- Stealing, breaking or damaging of computing devices

Malware Attack

- A generic term for software that has malicious purpose

Hacking Attack

- Any attempt to gain unauthorized access to your system

Denial of Service (DoS) Attack

- Blocking access from legitimate users

Computer security/ Attacks & Threats

Malware Attack: Examples

- Viruses

- Worms

- Trojan horses

- Spy-wares

- Spam/scam, identity theft, e-payment frauds, etc.

Computer security/Threats

Malware Attack ...

Viruses

- “A small program that replicates and hides itself inside other programs usually without your knowledge.” by Symantec
- Similar to biological virus: Replicates and Spreads by its own

Worms

- An independent program that reproduces by copying itself from one computer to another
- It can do as much harm as a virus
- It often creates denial of service

Computer security/Threats

Malware Attack...

Trojan horses

- (Ancient Greek tale of the city of Troy and the wooden horse) - ??
- Secretly downloading a virus or some other type of malware on to your computers.

Spy-wares

- “A software that literally spies on what you do on your computer.”
- Example: Simple Cookies and Key Loggers

Computer security/Threats

Most software based attacks are commonly called Viruses:
How do viruses work?

Infection mechanisms

- First, the virus should search for and detect objects to infect
- Installation into the infectable object
 - Writing on the boot sector
 - Add some code to executable programs
 - Add some code to initialization/auto-executable programs
 - Write a macro in a word file
 - ...

Computer security/Threats

How do viruses work? ...

☰ Trigger mechanism

- Date
- Number of infections
- First use

☰ Effects: **It can be anything**

- A message
- Deleting files
- Formatting disk
- Overloading processor/memory
- Etc.

Computer security/Threats

Who Writes Virus?

Adolescents

- Ethically normal and of average/above average intelligence.
- Tended to understand the difference between what is right and wrong
- Typically do not accept any responsibility for problems caused

Computer security/Threats

Who Writes Virus ...

College Students

- Ethically normal
- Despite expressing that what is illegal is “wrong”
- Are not typically concerned about the results of their actions related to their virus writing

Adults (smallest category)

- Ethically abnormal

Computer security/Threats

Anti-Virus

☰ There are

- Generic solutions
 - Ex. Integrity checking
- Virus specific solution
 - Ex. Looking for known viruses

☰ Three categories

- Scanners
- Activity monitors
- Change detection software

Computer security/Threats

Anti-Virus ...

Functions of anti-viruses

- Identification of known viruses
- Detection of suspected viruses
- Blocking of possible viruses
- Disinfection of infected objects
- Deletion and overwriting of infected objects

Computer security/Threats

Hacking /Intrusion/ Attack:

 **Hacking:** is any attempt to intrude or gain unauthorized access to your system either via some operating system flaw or other means. The purpose may or may not be for malicious purposes.

 **Cracking:** is hacking conducted for malicious purposes.

Computer security/Threats

Denial of Service (DoS) Attack:

☰ **DoS Attack:** is blocking access of legitimate users to a service.

☰ **Distributed DoS Attack:** is accomplished by tricking routers into attacking a target or using Zombie hosts to simultaneously attack a given target with large number of packets.